MILITARY SPECIFICATION

INDICATORS, POSITION, CONTROL SURFACES, 28 VOLT DC, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the general requirements for 28-V, d-c control surface electrical position indicators. The moving element is a nonenergized permanent magnet, and the stator is electrically connected to a position transmitter and energized with direct current.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-636	Boxes, Shipping, Fiberboard

Military

MIL-P-116	Preservation-Packaging, Method of
MIL-N-3336	Nut, Sheet Spring, Instrument Mounting
MIL-C-5015	Connector, Electric, AN Type
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-T-5583	Transmitter, Position, 28-Volt DC
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	Pointer, Dial, Standard Design of Aircraft Instrument

Air Force-Navy Aeronautical

AN5787	Transmitter-Position, Gearless

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be as specified herein and in accordance with the applicable detail specifications.

3.2 Qualification. Indicators furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.6).

3.3 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. Materials shall conform to applicable specifications and shall be as specified herein.

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3.4.1 Metals. Metals shall be of the corrosion-resistant type or suitably protected to resist corrosion during the normal service life of the indicator. Materials contained with hermetically sealed enclosures are considered to be suitably protected from corrosion. Requirements specified for fungicidal- and corrosion-protective treatment, anodizing of aluminum-alloy parts, and cadmium plating of steel parts are not applicable for parts contained within hermetically sealed indicators.

3.4.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator, except where magnetic materials are essential.

3.4.3 Fungusproof materials. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. In cases where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent as approved by the procuring activity.

3.4.4 Protective treatment. When materials are used in the construction of the indicator that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.5 Design and construction. The indicator shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service without failure. The indicator shall be so constructed as to permit overhaul and repair without requiring special tools and fixtures.

3.5.1 Case. The indicator case shall be in accordance with the detail specification and the following:

3.5.1.1 Body. The body of the case shall be made of suitable material, uniform in texture with a smooth surface. The bezel and that part of the case visible from the cockpit side of the instrument panel shall be finished in lusterless black, color No. 37038 of FED-STD-595. The remainder of the case may be finished in black and shall be adequately protected to satisfactorily withstand the environmental test conditions specified herein.

3.5.1.2 Reinforcement. The inside of the case around each connection shall be of sufficient strength to prevent damage to the case when the connections are tightened during installation of the indicator.

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3.5.1.3 Nonhermetically sealed cases. The body of nonhermetically sealed cases shall be made either of nonferrous, low-density metal or of synthetic materials, uniform in texture with a smooth surface, and entirely covered with a durable black finish. Synthetic material shall be of the thermal-setting type, composed of a filler and phenol composition product binder.

3.5.1.4 Hermetically sealed cases. The case shall provide a hermetically sealed enclosure for the internal mechanism of the indicator. The design of the case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed. This shall be accomplished without the use of special tools and fixtures unless they are approved by the procuring activity. Hermetic sealing shall be accomplished in such a manner as to make the seal independent of materials which will be affected by the action of any atmosphere to which the indicator may be subjected.

3.5.1.4.1 Filling medium. The hermetically sealed case shall be filled with a medium of 88 to 92 percent nitrogen and the remainder helium. The filling medium shall be of 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi. Other concentrations of helium shall not be used, except with prior approval of the procuring activity.

3.5.1.5 Bezel. When an adapter is furnished with an indicator to provide a removable bezel and mounting flange, the adapter shall be a fixed part of the case when installed. The bezel and mounting flange shall be made of nonferrous, low-density metal. The bezel and mounting flange shall have a durable, lusterless black finish.

3.5.1.5.1 Installation. When a screw-type mounting bezel is provided, the following shall apply:

3.5.1.5.1.1 Mounting screws. Sufficient mounting screws shall be provided for installing the indicator. They shall be No. 6-32, NC-2 round head, brass machine screws having a durable, lusterless black finish. The length of the screws shall permit mounting on a panel 3/16 inch thick.

3.5.1.5.1.2 Spring nuts. A sufficient number of removable No. 6-32, NC-2 spring nuts shall be provided to fit the mounting lugs of the indicator. The nuts shall be in accordance with MIL-N-3336.

3.5.1.5.1.3 Envelope. An envelope, containing the mounting screws and spring nuts, shall be provided with each indicator. The following information shall be provided on the face of the envelope:

IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS AND
SPRING NUTS

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3.5.1.6 Cover glass. The distance between the outer surface of the cover glass and the front edge of the case shall be the minimum practicable, but shall not exceed 1/32 inch. The cover glass shall be clear, flat, and free from flaws which interfere with the normal reading of the indicator. The thickness of the cover glass shall be the minimum practicable and entirely suitable for the purpose. The distance between the inner surface of the cover glass and the dial shall be the minimum practicable and shall not exceed 1/8 inch.

3.5.1.6.1 Fogging. There shall be no evidence of moisture or oil fogging of the cover glass when the indicator is subjected to a temperature of 85° ±5°C (185° ±9°F) for a minimum of 2 hours, followed immediately by immersing the face of the glass in an ice bath for 3 minutes.

3.5.2 Instrument dial. The instrument dial shall conform to the detail specification. The style and proportions of numerals and letters placed on the dial shall conform to MS33558. Numerals shall distinctly indicate the graduation to which each applies. All visible portions of the dial shall be lusterless black, color No. 37038 of FED-STD-595.

3.5.2.1 Markings. All markings shall be durable to withstand usage encountered in service. All markings on the dial shall be lusterless white, color No. 37875 of FED-STD-595.

3.5.3 Pointer. The pointer shall conform to MS33585. It shall be as light as practicable and sufficiently rigid to prevent oscillation.

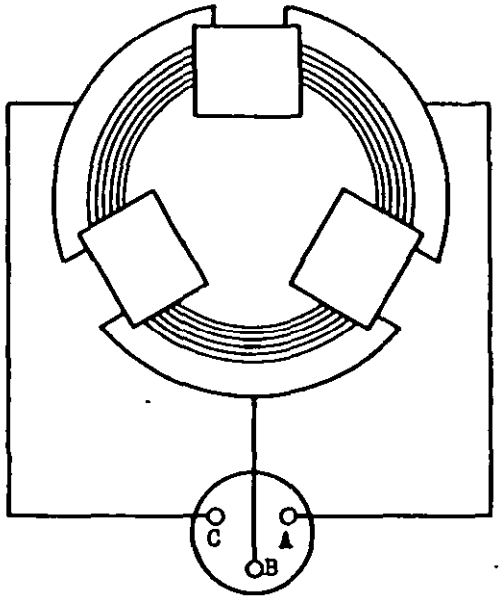
3.5.4 Electrical connectors. The electrical connectors used in the construction of the indicator shall be in accordance with MIL-C-5015, except in hermetically sealed indicators where a receptacle designed for hermetic sealing shall be used. The receptacle shall be suitably protected from corrosion. The polarizing key of the receptacle shall be located toward the top of the indicator. The pins shall be identified as shown on figure 1.

3.5.4.1 Wiring. The internal wiring to the electrical receptacle of the indicator shall be of flexible stranded wire and shall conform to figure 1. The external wiring connections shall be as shown on figure 1 for clockwise pointer travel as the transmitter shaft is rotated clockwise.

3.5.4.2 Zero reading. The indicator reading shall be 0 when the plus brush of the transmitter is over B tap. The transmitter wiring shall be as shown on figure 1 of MIL-T-5583.

3.5.5 Screw threads. Unless otherwise specified, the threads of all machine screws, 0.060 inch or larger in diameter, shall conform to MIL-S-7742.

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INTERNAL WIRING - INDICATOR

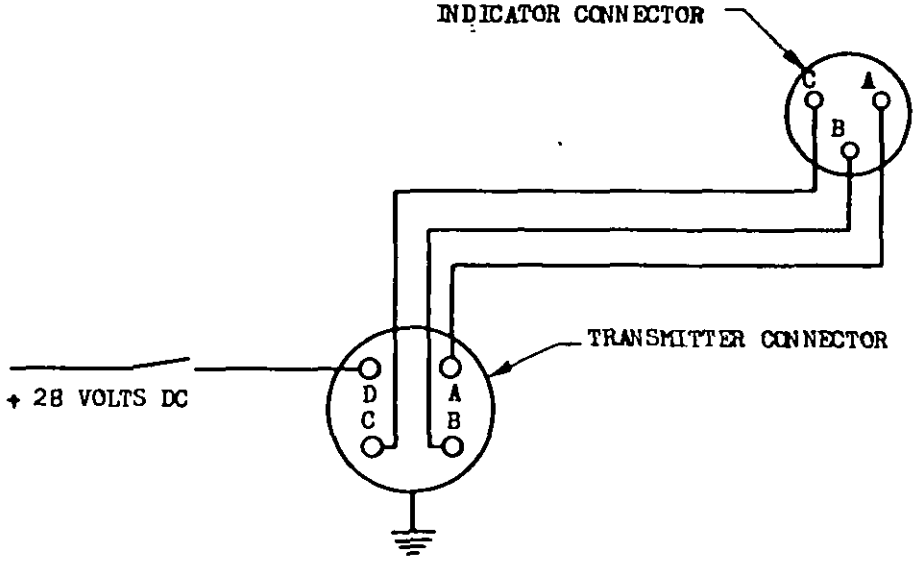


FIGURE 1. Wiring diagram

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3.6 Performance. The indicator shall meet the performance requirements specified in section 4 when subjected to the applicable tests.

3.7 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.8 Weight. The weight of the indicator completely assembled, less mounting nuts and mounting screws, shall not exceed 0.5 pound.

3.9 Finishes and protective coatings

3.9.1 Aluminum-alloy parts. Aluminum-alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

3.9.1.1 Dials, small holes, and case inserts need not be anodized.

3.9.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.9.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.9.1.4 Castings containing nonaluminum-alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.9.1.5 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.9.2 Steel parts. Steel parts, except stainless steel, shall be cadimum plated, where practicable, in accordance with QQ-P-416, types II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.10 Identification of product

3.10.1 Nameplate. A nameplate, permanently, and legibly marked with the information specified in the detail specification, shall be securely attached to the exterior of the indicator case. The information marked in the spaces provided on the nameplate shall be in accordance with MIL-STD-130.

3.10.2 Use of AN or MIL designation. AN or MIL designations shall not be applied to a product, except for qualification test samples, nor referred to in correspondence, until notice of approval has been received from the activity responsible for qualification.

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3.11 Workmanship. The indicator, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and assemblies, and freedom of parts from burrs and sharp edges.

3.11.1 Dimensional tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.11.2 Screw assemblies. Assembled screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further, without damage or injury to the screw, bolt, or threads.

3.11.3 Cleaning. The indicator shall be thoroughly cleaned of loose, spattered, or excess solder; metal chips; and other foreign material after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the indicator shall be classified as follows:

a. Qualification testing . . . . . See 4.4

b. Quality conformance inspection . . . . . See 4.5.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 77°F) (25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowances shall be made for the difference from the specified condition.

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4.3.2 Attitude. Unless otherwise specified, the indicator shall be tested in its normal operating position.

4.3.3 Tapping. Unless otherwise specified, the indicator shall be gently tapped or vibrated before a test reading is taken. Vibration shall be applied in the axis perpendicular to the dial at 60 Hz with a maximum ampliture of 0.004 inch.

4.3.4 Vibration stand. A vibration stand shall be used which will vibrate at any desired frequency between 8 and 50 Hz and subject the indicator to vibration such that a point on the case will describe in a plane inclined 45 degrees to the horizontal plane, a circle, the diameter of which is equal to the amplitude specified herein.

4.3.5 Test transmitter. When a test transmitter is specified in the tests, it shall conform to MIL-T-5583 and its scale errors shall be known in terms of the average production of the manufacturer; or a device shall be used which is capable of simulating an equivalent electrical circuit. Readings obtained shall be corrected for scale errors of the test transmitter.

4.4 Qualification testing

4.4.1 Test samples. The test samples shall consist of two complete indicators and two indicator cases representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the qualifying activity.

4.4.2 Test report. When tests are conducted at a location other than the laboratory of the qualifying activity, two copies of a test report in accordance with MIL-STD-831 shall be furnished to that activity.

4.4.3 Qualification tests. The qualification tests shall consist of all the tests specified under 4.6.

4.5 Quality conformance inspection. The quality conformance inspection shall consist of the following:

a. Individual tests . . . . . See 4.5.1

b. Sampling plan and tests . . . . . See 4.5.2.

4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6:

a. Examination of product . . . . . See 4.6.1

b. Voltage off test . . . . . See 4.6.2

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c. Operation test . . . . . See 4.6.3

d. Scale error test . . . . . See 4.6.4

e. Friction test . . . . . See 4.6.5

f. Position error test . . . . . See 4.6.6

g. Insulation leakage test . . . . . See 4.6.7

h. Sealing test . . . . . See 4.6.22.

4.5.2 Sampling plan and tests. Five indicators selected from each five hundred or fraction thereof on the contract or order shall be subjected to the following tests as described under 4.6:

a. Low temperature operation test . . . . See 4.6.8

b. High temperature operation test . . . See 4.6.9

c. Magnetic effect test . . . . . See 4.6.10

d. Vibration error test . . . . . See 4.6.11

e. Voltage variation test . . . . . See 4.6.12

f. Humidity test . . . . . See 4.6.15

g. Salt-fog test . . . . . See 4.6.16

h. Indicator cycling test . . . . . See 4.6.21

i. Fogging test . . . . . See 4.6.24.

4.5.2.1 Rejection and retest. When an indicator selected from a production run fails to meet the specification, items still on hand or later produced shall not be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain fully to the Government representative and notify the procuring activity in writing the cause of failure, the action taken to preclude recurrence, and the impact this failure may have on scheduled deliveries.

4.5.2.2 Individual tests may continue. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or items produced later shall not be made until it is determined that all items meet all the requirements of the specification.

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4.5.2.3 Defects in indicators already accepted. The investigation of a test failure could indicate that defects may exist in indicators already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and the method of correcting them.

4.5.3 Special tests. Three out of the first fifteen indicators or less submitted for acceptance under contract shall be selected at random and subjected to the following tests:

NOTE: The tests identified herein as special tests are not required as part of quality conformance inspection, but shall be performed during qualification inspection only.

a. Dust test . . . . . See 4.6.13

b. Rain test . . . . . See 4.6.14

c. Extreme low temperature exposure test . . . . . See 4.6.17

d. High temperature exposure test . . . . . See 4.6.18

e. Fungus resistance test . . . . . See 4.6.19

f. Vibration failure test . . . . . See 4.6.20

g. Thermal shock cycling test . . . . . See 4.6.23.

4.5.4 Rejection and retest. Indicators which have been rejected may be reworked or replaced to correct the defects, and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the procuring activity. Units rejected after retest shall not be resubmitted without specific approval of the procuring activity.

4.6 Test methods

4.6.1 Examination of product. The indicator shall be inspected to determine compliance with this specification with respect to materials, workmanship, and envelope.

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4.6.2 Voltage off test. With no voltage applied to the indicator, the pointer shall move off scale from the zero graduation to a position that is at or near the center of the off-scale portion of the dial, provided the total distance to the off-scale pointer rest position is not less than 20 or more than 50 mechanical degrees.

4.6.3 Operation test. The indicator and transmitter shall be connected to a 28-V d-c source as shown on figure 1. The pointer shall move smoothly and uniformly throughout its range of travel as the transmitter shaft is rotated.

4.6.4 Scale error test. The transmitter shall be so positioned that the indicator pointer coincides with each major dial graduation. The indicator shall be lightly tapped while making these settings. The difference between the correct and the actual transmitter readings is the scale error and shall be recorded. The scale error shall not exceed 6 mechanical degrees.

4.6.5 Friction test. The transmitter shall be so positioned that the indicator pointer coincides with each major dial graduation. Two readings shall be taken, one before and one after the instrument is tapped. The difference between these two readings is the friction error and shall be recorded. The friction error shall not exceed 3 mechanical degrees. This test may be combined with the scale error test.

4.6.6 Position error test. The indicator shall be tested at the 20-degree indication or as specified by the procuring activity. The change in pointer indication produced by rotating the indicator from the normal operating position as follows shall not exceed 2 mechanical degrees:

a. Rotated 90 degrees back, the dial horizontal and face up
b. Rotated 90 degrees to the right, the dial in a vertical plane
c. Rotated 90 degrees to the left, the dial in a vertical plane.

4.6.7 Insulation leakage test. The potential of 100V, at commercial frequency or direct current, shall be applied between any electrical receptacle pin contact and the shell, shielding, or any metal part of the indicator case. The insulation resistance measured between any pin and shell, shielding, or any metal part of the indicator shall be a minimum resistance of 20 megohms.

4.6.8 Low temperature operation test. The indicator shall be subjected to a temperature of -65° ±2°F (-54° ±2°C) for a period of 4 hours. At the end of this period and while still at the low temperature, the indicator shall be subjected to the scale and friction error tests. The scale error shall not exceed 6 mechanical degrees, and the friction error shall not exceed 4 mechanical degrees.

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4.6.9 High temperature operation test. The indicator shall be placed in a chamber and maintained at a temperature of 160° ±2°F (71° ±2°C) for a period of 4 hours. While still at this temperature, the indicator shall be subjected to the scale and friction error tests. The scale error shall not exceed 6 mechanical degrees, and the friction error shall not exceed 4 mechanical degrees. After the indicator has returned to room temperature, it shall meet the test specified in 4.6.4.

4.6.10 Magnetic effect test. The indicator, first not operating and second operating on rated voltage, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0, 45, 90, 135, 180, 225, 270, and 315 degrees from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in its normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 gauss. The deflection of the compass at any of the specified positions shall not exceed 2.5 degrees.

4.6.11 Vibration error test. The indicator shall be so mounted on a vibration stand (see 4.3.4) that the dial is in the vertical plane and the dial markings are in the normal operating position. The indicator shall be energized, and the pointer set to various indications during the test. The indicator shall be subjected to vibration with an amplitude between 0.009 and 0.011 inch at frequencies of from 8 to 50 Hz for a period of 4 hours.

4.6.11.1 Pointer oscillation. The maximum amplitude of the pointer oscillation during the vibration error test shall not exceed ±3 mechanical degrees.

4.6.11.2 Pointer variation. The reading shall not differ at any time by more than 3 mechanical degrees from the reading obtained when the indicator is at rest.

4.6.12 Voltage variation test. The indicator shall be operated at a reduced voltage of 22-1/2V and an increased voltage of 30V at an indication of 25 degrees, or as specified by the procuring activity. The indication shall not change more than 2.5 mechanical degrees between operation at the reduced voltage and operation at the increased voltage. The indicator shall be tapped lightly after each change in voltage.

4.6.13 Dust test. This test is applicable only to nonhermetically sealed indicators. The indicator shall be subjected to a dust test in accordance with MIL-STD-810, Method 510, Procedure I. At the completion of the test, the indicator shall be removed and allowed to cool to room temperature. The indicator shall then be subjected to the scale error and friction tests. The scale error and friction errors shall not exceed the tolerance allowed at room temperature. The indicator shall be carefully examined to ascertain that no sand has entered the interior of the case.

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4.6.14 Rain test. This test is applicable only to nonhermetically sealed indicators. The indicator shall be subjected to a rain test in accordance with MIL-STD-810, Method 506, Procedure I.

4.6.15 Humidity test. This test is applicable only to nonhermetically sealed indicators. The indicator shall be subjected to a humidity test in accordance with MIL-STD-810, Method 507, Procedure I. External connections shall be made to the indicator to simulate installed conditions. Immediately following the completion of 15 cycles, the indicator shall be subjected to the tests specified in 4.6.7, and the measured resistance shall be a minimum of 20 megohms. The indicator shall then be subjected to the scale error and friction tests. The room temperature scale and friction error tolerances shall apply. The indicator shall be examined internally and externally. There shall be no evidence of corrosion or deterioration which would affect subsequent operation.

4.6.16 Salt-fog test. The indicator shall be subjected to a salt-fog test in accordance with MIL-STD-810, Method 509, Procedure I for a period of 50 hours. External connections shall be made to the indicator in such a manner as to simulate installed conditions. At the end of the 50-hour period, the indicator shall be subjected to the scale error and friction tests. The room temperature scale and friction error tolerance shall apply.

4.6.17 Extreme low temperature exposure test. The indicator shall be subjected to a temperature of -62° ±2°C (-80° ±4°F) for a period of 48 hours. At the end of the 48-hour period and while still at the low temperature, the indicator shall be operated throughout its range without exhibiting sticking or erratic characteristics. The temperature shall then be raised to -54° ±2°C (-65° ±4°F) and maintained for an additional 24-hour period. At the end of the 24-hour period and while still at the low temperature, the indicator shall then be subjected to the scale error and friction tests. The low temperature scale and friction error tolerances shall apply, and there shall be no damage to the indicator which would adversely affect subsequent operation.

4.6.18 High temperature exposure test. The indicator shall be subjected to a temperature of 71° ±2°C (160° ±4°F) for a period of 24 hours. At the end of the 24-hour period and while still at the high temperature, the indicator shall be tested at this temperature for scale errors and friction as specified in the individual tests. The high temperature operation, scale error, and friction tolerances shall apply. Then the indicator shall be examined for deterioration or damage due to extreme high temperature exposure.

4.6.19 Fungus resistance test. This test is applicable only to nonhermetically sealed indicators. The indicator shall be subjected to the action of fungi in accordance with MIL-STD-810, Method 508, Procedure I. The unit selected shall not have previously been subjected to the salt spray test. At the end of the test period, the indicator shall be carefully examined to ascertain that no fungus growth has occurred. At the option of the procuring activity, tests in addition to or in lieu of the above may be made on individual components of the indicator.

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4.6.20 Vibration failure test. The test shall be conducted while the indicator is being operated on 28V dc. The indicator shall be subjected to vibration with a total excursion or double amplitude between 0.018 and 0.020 inch for a 3-hour period. The frequency shall be varied uniformly from 8 to 50Hz and returned once each hour during this test. After the vibration, the indicator shall be subjected to the scale error and position error tests. The tolerances for scale and position errors at room temperature shall apply. No looseness in the mechanism nor damage to any part of the indicator shall result from the vibration.

4.6.21 Indicator cycling test. The indicator shall be operated approximately over 90 percent of the scale and back not less than 20,000 cycles. No damage to the indicator shall result from this test. The indicator shall be tested for scale and friction error. The final error shall not exceed 6 mechanical degrees, and the friction error shall not exceed 3.5 mechanical degrees.

4.6.22 Sealing test (hermetically sealed indicators only). The indicator case shall be tested for leaks by means of a mass-spectrometer-type helium leak detector. The detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 12 to 14.7 psi.

4.6.23 Thermal shock cycling test. This test is applicable only to hermetically sealed indicators. The indicator shall be alternately subjected to a total of 8 cycles of immersion in tap water maintained at 185° ±5°F and 41° ±5°F (85° ±5°C and 5° ±5°C). The length of time for each bath immersion shall be 30 minutes; not more than 5 seconds shall elapse between bath immersions. No evidence of moisture penetration or damage to the enclosure shall result from this test. Following this test, the indicator shall be resubjected to the test specified in 4.6.22. No indicator leakage shall occur as a result of this test.

4.6.24 Fogging test. This test is applicable only to hermetically sealed indicators. The indicator shall be placed in a controlled ambient temperature of 185° ±5°F (85° ±5°C) for a minimum of 2 hours. Within 5 seconds after removal from the oven, the indicator face shall be immersed to a depth of 1/4 to 1/3 inch in an ice bath maintained at 32° to 41°F (0° to 5°C). After 3 minutes, the indicator shall be removed from the ice bath and quickly wiped dry (compressed air shall not be used). The cover glass and dial shall immediately be inspected for evidence of fogging or damage. Evidence of fogging or damage shall be cause for rejection.

4.7 Inspection of the preservation, packaging, packing, and marking for shipment and storage. The inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Indicators shall be preserved and packaged in accordance with MIL-STD-794 and Method III of MIL-P-116, one each in a unit container conforming to PPP-B-636, domestic class. Electrical receptacles and connectors shall be equipped with water-vaporproof caps. If not hermetically sealed, preservation method shall be selected from the table in MIL-STD-794.

5.1.2 Level C. Indicators shall be individually preserved and packaged in a manner that will provide protection against corrosion, deterioration, or physical damage so that serviceability is assured during shipment from supply source to the first receiving activity. This level may be the contractor's commercial practice provided the latter meets the criteria for this level as defined in MIL-STD-794.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Indicators preserved and packaged as specified in 5.1.1 shall be packed as required for consolidation in shipping boxes conforming to PPP-B-636, weather-resistant class. Exterior shipping containers shall be of uniform shape and size, of minimum tare, weight, and cube.

5.2.2 Level B. Indicators preserved and packaged as specified in 5.1.1 shall be packed as required for consolidation in accordance with PPP-B-636, domestic class exterior shipping containers. Other requirements as specified in 5.2.1 shall apply.

5.2.3 Level C. Indicators shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery, and shall meet as a minimum, the requirements of carrier rules and regulations. This level may be the contractor's commercial practice provided the latter meets the criteria for this level as defined in MIL-STD-794.

5.3 Marking. In addition to the standard marking requirements of MIL-STD-129, level C/C packs containing item(s)/equipment intended for immediate use at destination shall be conspicuously marked as follows on two opposite sides of the exterior container in letters as large as space permits:

FOR IMMEDIATE USE

DO NOT STORE IN THIS PACK

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6. NOTES

6.1 Intended use. The position indicators covered by this specification are intended to indicate the variation from their neutral position of aircraft components such as rudder, elevator, aileron trim tabs, flaps, and nose steering wheel.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Required levels of preservation and packaging, and packing (see section 5).

6.3 Data. Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) referencing the appropriate data item description in the military departments' Authorized Data List (ADL). The data produced by this specification is the test report specified in 4.4.2.

6.4 Definitions. For the purpose of this specification, the following definition shall apply:

6.4.1 Amplitude. Amplitude is defined as the extent or motion as measured from one extreme to the opposite extreme.

6.4.3 Hermetic seal. A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic materials. A hermetic seal is not intended to include seals accomplished by gaskets.

6.5 Identification of changes. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.6 Qualification. With respect to products requiring qualification, awards will be made only for such products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products Lists is the Aeronautical Systems Division, Attn: ASD/ENFIP, Wright-Patterson Air Force Base, Ohio 45433, and information pertaining to qualification of products may be obtained from that activity.

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